atoms were bombarded by alpha rays. The nucleus of a hydrogen atom is an elementary particle, and as we shall see later, it is—for this very reason—one of the most important fundamental building blocks of matter. Consequently, it has been named *proton.* When nitrogen atoms are bombarded by alpha particles—in other words, by helium nuclei—a proton is emitted occasionally by the nitrogen nucleus. The alpha particle remains in the nucleus. The laws of the conservation of mass and energy permit us to calculate what happens to the nucleus in such a case. The nitrogen nucleus has the mass number 14 and the nuclear charge number (atomic number) 7; its symbol, therefore, is ${}_7N^{14}$. The mass number and charge number of the alpha particle are, respectively, 4 and 2, and its symbol is ${}_2He^4$, as already mentioned; the symbol of the proton is ${}_1H^1$. As regards mass, the nitrogen nucleus is changed by the absorption of an alpha particle and the loss of a proton, as follows:

$$14 + 4 - 1 = 17$$

But as regards its charge, the following equation applies:

$$7 + 2 - 1 = 8$$

Thus, a nucleus of mass 17 and nuclear charge (atomic number) 8 is formed. This nuclear charge number indicates that it is an oxygen atom, but the mass number 17 does not agree with the mass number of the common oxygen atom, which is 16. The truth is that this is a rare *isotope* of oxygen. We shall discuss such isotopes in due course. This transformation of ${}_7N^{14}$ into ${}_8O^{17}$ is the renowned first instance of the artificial transmutation of elements.

Such nuclear reactions can be represented by appropriate formulae. The formula of this particular reaction is:

$$_7N^{14} + {}_2He^4 \rightarrow {}_8O^{17} + {}_1H^1$$

Such nuclear transformations can be observed in the cloud chamber (*Blackett*). But since these are very rare phenomena, many thousands of